

Networks and Security

Alireza Esfahani *BSc, MSc, PhD, FHEA, M IEEE, M ECSO*
Senior Lecturer in Cyber Security
Cyber Security Course Leader
School of Engineering and Computing
University of West London



UNIVERSITY OF
WEST LONDON

The *Career* University

Setting Up a Wireless Network

Wireless modules

- ❑ **Linksys-WMP300N:** This is available for servers, PCs, and laptops. It provides one 2.4 GHz wireless interface with protocols supporting Ethernet. Once plugged in, this module is configurable through the PC Wireless utility available in the Desktop tab.
- ❑ **PC-HOST-NM-1W:** This is a basic wireless interface that provides one 2.4 GHz wireless interface with support for Ethernet. This module isn't configurable.
- ❑ **PC-HOST-NM-1W-A:** This is similar in features to the preceding module, except that it provides a 5 GHz wireless interface.
- ❑ **HWIC-AP-AG-B:** This is a router module that works with 1841 and 2811 routers. It functions as an integrated access point and supports single band 802.11b/g or dual band 802.11a/b/g radios.

Wireless modules (cont.)

- ❑ **TabletPC-PT / PDA-PT / WirelessEndDevice-PT:** These three devices provide the same functionality; they only represent different things. These devices come with a built-in wireless interface.
- ❑ **AccessPoint-PT / AccessPoint-PT-A / AccessPoint-PT-N:** These are generic wireless access points with minimal configuration options. All of them have an antenna and a port to connect to a device, such as a router or DHCP server.
- ❑ **Linksys-WRT300N:** This wireless device provides a web interface similar to the one provided by a real Linksys model for configuring a SSID (Service Set Identifier), wireless authentication, WAN port, and much more. This device has 4 Ethernet LAN ports and 1 Ethernet WAN port which can be connected to a router or model that provides internet.

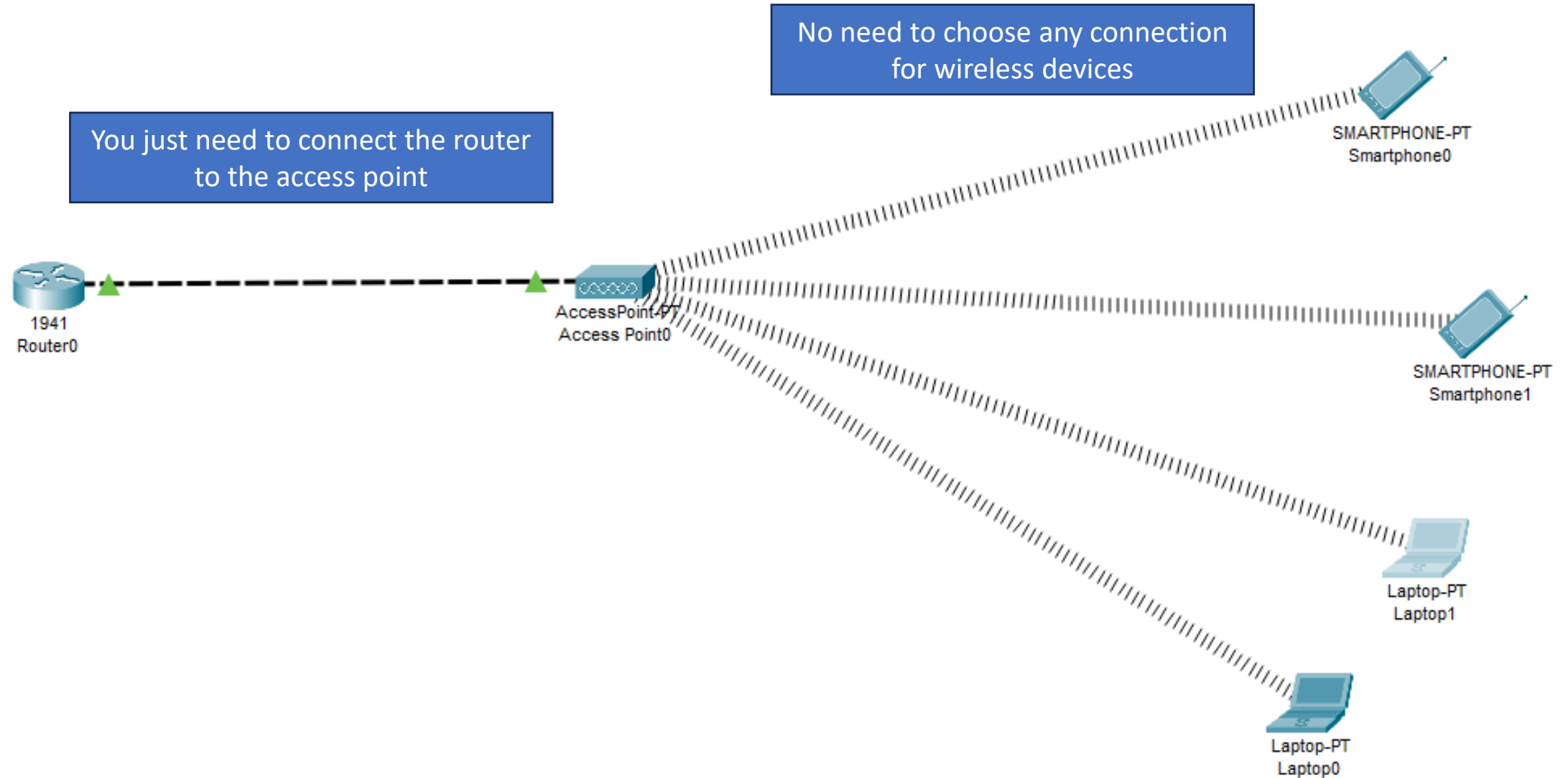
Exercise

Introduction

First, open the Cisco packet tracer desktop and select the devices given below:

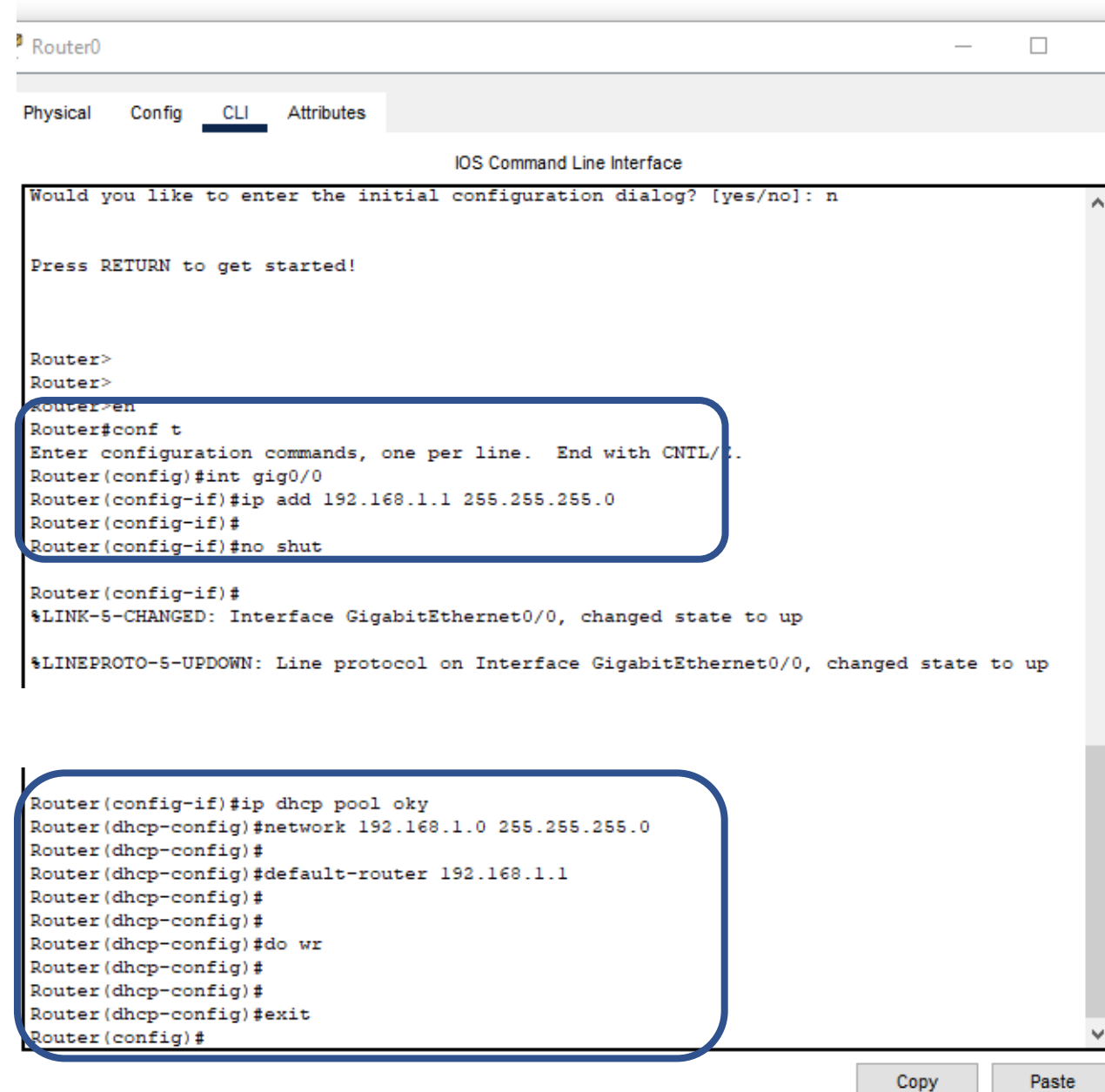
Device	Model Name	Quantity
Router	Router 1941	1
Smartphone	Smartphone	2
Laptop	Laptop	2
Access Point	AP-PT	1

Topology



Router configuration

- For the Router, configure it based on the given Figure.
- The format of the command is as follows:
- *ip dhcp pool <pool-name>*



The screenshot shows the Router0 CLI interface with the 'CLI' tab selected. The interface displays the following commands and their outputs:

```
Router0
Physical Config CLI Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shut
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config-if)#ip dhcp pool oky
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#do wr
Router(dhcp-config)#
Router(dhcp-config)#
Router(dhcp-config)#exit
Router(config)#
```

At the bottom right of the interface, there are 'Copy' and 'Paste' buttons.

Access Point configuration

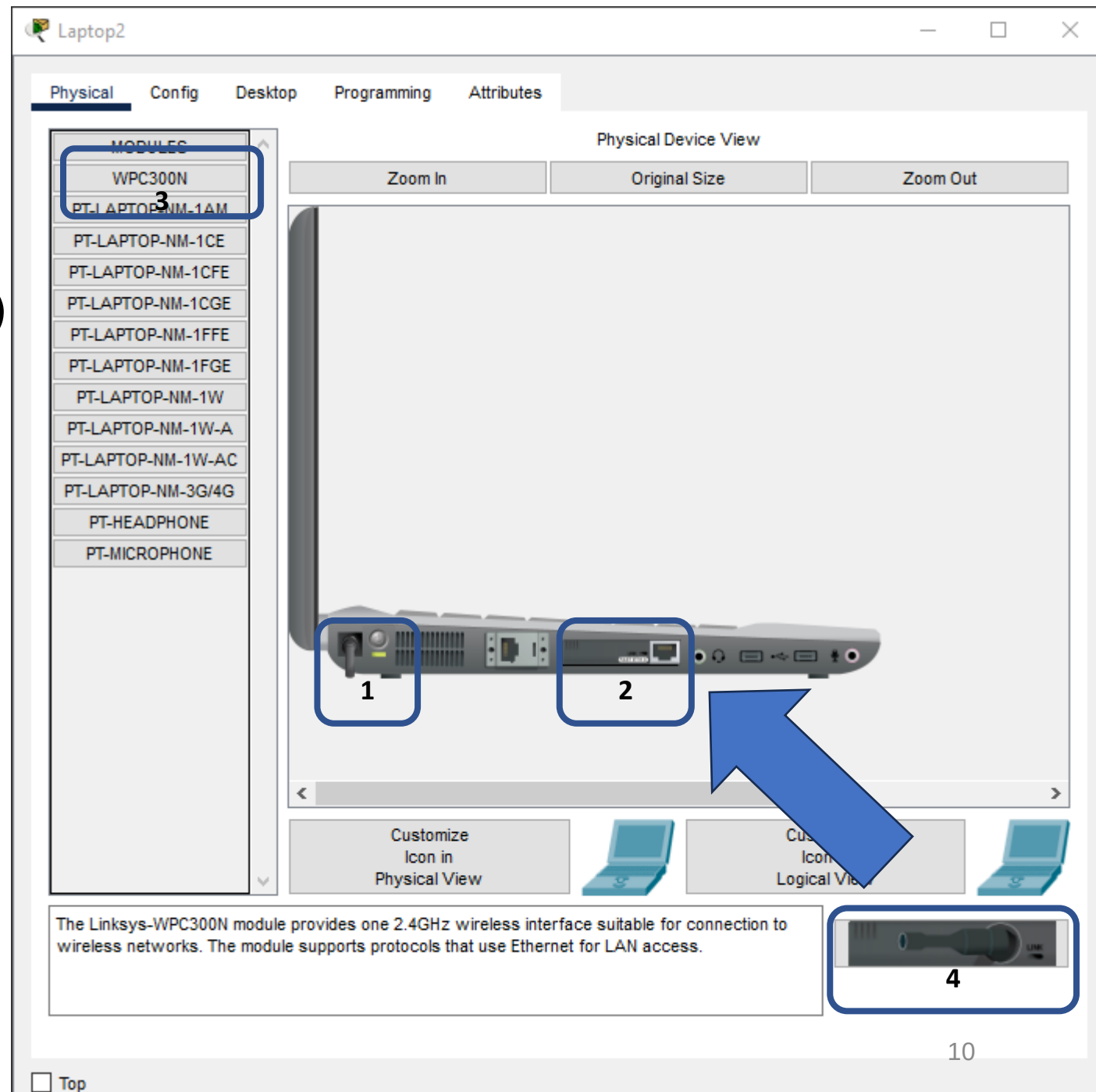
- Give a name for SSID
- Use WPA2-PSK and set up your password

The screenshot shows the 'Access Point0' configuration window with the 'Config' tab selected. The left sidebar shows a tree view with 'GLOBAL' and 'INTERFACE' sections. Under 'INTERFACE', 'Port 0' and 'Port 1' are listed, with 'Port 1' selected. The main area displays the configuration for 'Port 1'. The 'Port Status' is 'On'. The 'SSID' is 'CP50088'. The '2.4 GHz Channel' is set to '1'. The 'Coverage Range (meters)' is '140.00'. Under 'Authentication', 'WPA2-PSK' is selected. The 'WEP Key' is empty. The 'PSK Pass Phrase' is '123456789'. The 'User ID' is empty. The 'Password' is empty. The 'Encryption Type' is 'AES'.

Port 1	
Port Status	☒ On
SSID	CP50088
2.4 GHz Channel	1
Coverage Range (meters)	140.00
Authentication	
☐ Disabled	☐ WEP
☐ WPA-PSK	☒ WPA2-PSK
WEP Key	
PSK Pass Phrase	123456789
User ID	
Password	
Encryption Type	AES

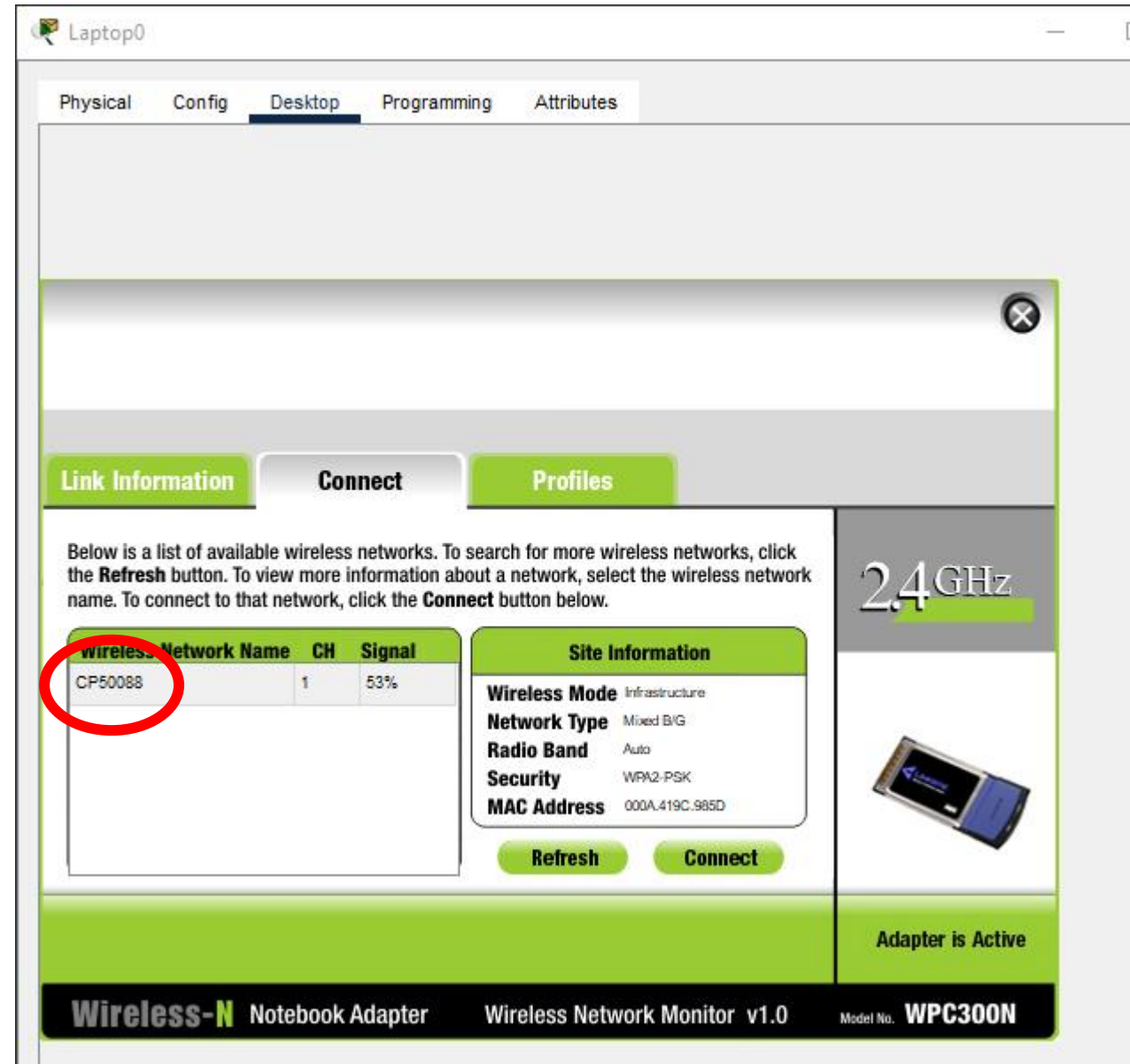
Laptop configuration

- Power off the laptop (Step 1)
- Remove the Ethernet card (Step 2)
- Install the WPC300N card (Step 3)
- Then replace it (Steps 4).



Laptop configuration (cont.)

- Navigate to Desktop > PC Wireless.
- Locate the SSID network assigned to the access point.
- Connect to the access point.
- Enter the password you set for the access point.



Smartphone configuration

- Use the same setting

The image shows a configuration window for a device named 'Smartphone0'. The window has several tabs: 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Config' tab is selected. On the left side of the 'Config' tab, there is a tree view with the following structure:

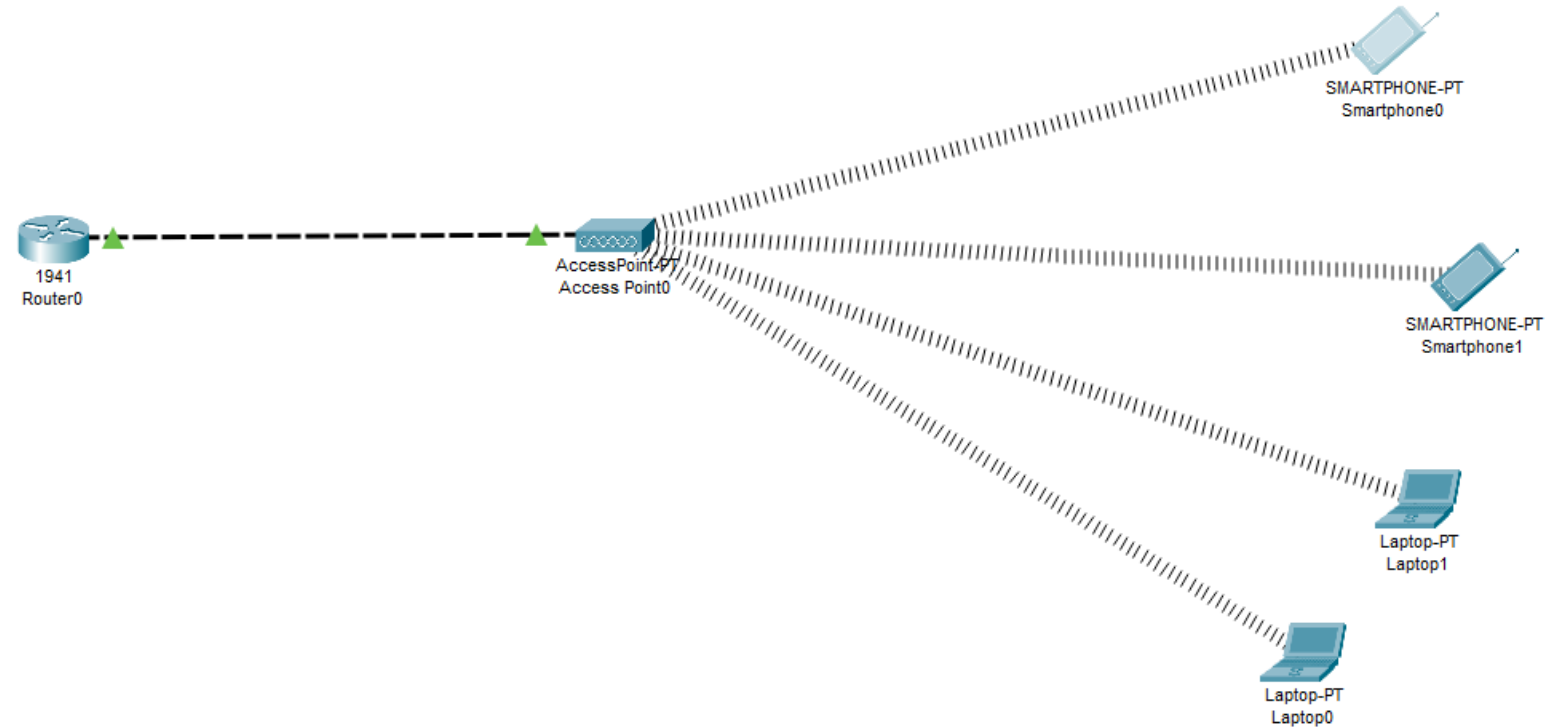
- GLOBAL
 - Settings
 - Algorithm Settings
- INTERFACE
 - Wireless0 (selected)
 - 3G/4G Cell1
 - Bluetooth

The main area of the window displays the configuration for 'Wireless0'. The settings are as follows:

- Port Status: ☒ On
- Bandwidth: 54 Mbps
- MAC Address: 0002.16C2.88C5
- SSID: CP50088
- Authentication:
 - ☐ Disabled
 - ☐ WEP
 - ☐ WPA-PSK
 - ☒ WPA2-PSK
 - ☐ WPA
 - ☐ WPA2
 - ☐ 802.1X
 - Method:
- WEP Key:
- PSK Pass Phrase: 123456789
- User ID:
- Password:
- MD5 (dropdown menu)
- User Name:
- Password:
- AES (dropdown menu)
- Encryption Type:
- IP Configuration:
 - ☒ DHCP
 - ☐ Static
- IPv4 Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- IPv6 Configuration:
 - ☒ Automatic
 - ☐ Static
- IPv6 Address:
- Link Local Address: FE80::202:16FF:FEC2:88C5

Outcome

- Verify if you can ping each device.



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delet
	Successful	Laptop0	Smartphone0	ICMP		0.000	N	0	(edit)	
	Successful	Laptop0	Smartphone1	ICMP		0.000	N	1	(edit)	